LessonPlan Career Roadmap

Polytechnic Diploma and ITI (Technical Careers after Class 8 or 10): Class 6 to Career

STATUS: DRAFT FOR HUMAN REVIEW - NOT FOR PUBLIC USE
Drafted and web-checked on 17 Oct 2026 by an AI assistant. Karnataka-specific 2026 notifications for diploma and ITI admission were not found; most facts below come from aggregator and college-listing pages and general national guidance. Treat every item as needing confirmation on dtetech.karnataka.gov.in, the Directorate of Employment and Training (ITI) and the KEA site before publishing. See section 9 for the verification log.

1. Pathway overview

Stage	When	Key decision	What it unlocks
Foundation	Class 6-8	Build hands-on skills, Maths and Science basics; try workshops, tinkering and craft	Technical aptitude
Class 8 entry	After Class 8	Some ITI trades accept Class 8 pass (Welder, Plumber, Carpenter, Mason and similar)	NCVT trade certificate, apprenticeship
Class 10 entry	After Class 10 (SSLC)	Polytechnic diploma (3 years) or ITI trade (6 months to 2 years)	Job-ready technical skills; lateral entry to engineering
Alternative	After Class 10	Continue to Class 11-12 for degrees	Compare time, cost and goal with a diploma
Diploma to degree	After diploma	Lateral entry into 2nd year BE / BTech through DCET (Karnataka)	Engineering degree in 3 more years
Job / apprenticeship	After ITI / diploma	Apprenticeship at PSUs, railways and industry; private and government technician jobs	Employment and certification
Career	-	Technician, junior engineer, supervisor, entrepreneur	See section 7

2. Stage-by-stage nodes

Class 6-10: Foundation

- Decision: develop practical skills (electrical basics, tools, drawing, computing) alongside Maths and Science.
- Keep Maths and Science marks decent; polytechnic admission in Karnataka reportedly needs a Class 10 pass with Science and Maths **[VERIFY minimum marks]**.

After Class 8

- ITI: Class 8 pass is enough for some trades (Welder, Plumber, Carpenter, Mason, Painter, Mechanic Agriculture Machinery). **[VERIFY Karnataka trade list]**

After Class 10 (SSLC)

- Polytechnic diploma: 3 years in Mechanical, Civil, Electrical (EEE), Electronics (ECE), Computer Science, Automobile and other branches; admission coordinated by the Directorate of Technical Education (DTE) Karnataka. One source says merit on SSLC marks; another says entrance-based; confirm the process. **[VERIFY]**
- ITI: Class 10 pass is needed for trades like Electrician, Fitter, Turner, Machinist, Draughtsman, COPA and Stenography. Duration 6 months to 2 years; over 130 trades nationally.

After diploma or ITI

- Diploma holders can enter engineering degrees in the second year (lateral entry) via Karnataka DCET. ITI holders may also join the second-year diploma at some polytechnics. **[VERIFY DCET eligibility]**
- Apprenticeship: ITI is the standard route to apprenticeship at PSUs, Indian Railways and large manufacturers. **[VERIFY NAPS details]**

Career

- Technician, junior engineer, supervisor, CNC / machine operator, electrician, solar and EV technician, government technical posts; many start their own workshops.

3. Early preparation and school-level routes

Exam / route	Conducted by	Classes	Details	Timing
Tinkering labs, science exhibitions, workshops	Schools, NGOs, Atal Tinkering Labs **[VERIFY]**	6-12	Builds hands-on skill	Through the year
NMMS (scholarship test)	Karnataka DSERT / KSEAB	8	Single written test	Apply Sep-Oct; exam Jan-Feb

Tip for parents: a diploma or ITI is a strong, early-earning path. Compare it with Class 11-12 plus engineering, and remember lateral entry keeps the engineering-degree option open.

4. Admission routes and exams

Diploma (Polytechnic) admission - Karnataka	
Conducting body	Directorate of Technical Education (DTE) Karnataka; dtetech.karnataka.gov.in
Eligibility	SSLC (Class 10) pass; one listing says Science and Maths with at least 35% **[VERIFY]**
Selection	Centralised counselling; reported as SSLC merit-based by one listing and entrance-based by another **[VERIFY]**
Duration	3 years (6 semesters); lateral entry for ITI holders to 2nd year at some polytechnics **[VERIFY]**

DCET (Diploma CET) - lateral entry to engineering	
Conducting body	Karnataka Examinations Authority (KEA); cetonline.karnataka.gov.in/kea
Used for	Direct entry to the 2nd year of BE / BTech for diploma holders (and BSc graduates in some cases) **[VERIFY]**
Frequency / dates	Once a year; 2026 reported as applications 17 March - 3 April, exam 23 May **[VERIFY]**
Pattern	Not confirmed **[VERIFY]**

ITI (Industrial Training Institute) admission	
Eligibility	By trade: Class 8 pass for some trades, Class 10 pass for others; age norms per NCVT **[VERIFY]**
Selection	Mostly merit on Class 10 / 8 marks through state counselling; a few states use an ITI CET. Karnataka 2026 notification not found **[VERIFY]**
Duration	6 months to 2 years; National Trade Certificate (NTC) after the All India Trade Test (AITT) by NCVT
Apprenticeship	Standard route for PSU, railway and industry apprenticeship **[VERIFY]**

Other routes

Route	For	Key details
Apprenticeship (Apprentices Act / NAPS)	ITI and diploma holders	Paid on-the-job training with industry; details not confirmed **[VERIFY]**
Karnataka Skill Development Corporation (KSDC) short courses	After Class 10 / 12	See Skills and Entrepreneurship document **[VERIFY]**
Direct polytechnic / ITI-to-degree bridge courses	Diploma and ITI holders	Lateral entry rules change; check each university **[VERIFY]**

Postgraduate and beyond (preview)

Exam	Body	For	Tentative timing
GATE / university tests after BE	IIT-organised / universities	MTech after the degree	Feb **[VERIFY]**
Government technical posts (state PSC, SSC JE, RRB)	State / central	Junior Engineer and technician jobs	Per notification **[VERIFY]**

5. Scholarships and financial support

Scholarship	Provider	Who can apply	Amount	Window
State scholarships for diploma and ITI students	Karnataka SSP and departments	Category and income based	Varies	Typically Jul-Nov **[VERIFY]**
NMMS	Govt. of India via Karnataka DSERT	Class 8 in govt/aided/local body schools; family income up to 3.5 lakh; at least 55% in Class 7 (50% SC/ST) **[VERIFY: single source]**	12,000 per year (1,000 per month), Class 9-12	Apply Sep-Oct; exam Jan-Feb
CSSS (Central Sector Scheme)	Ministry of Education via National Scholarship Portal	Top 20% of successful candidates in own board and stream; family income up to 4.5 lakh	12,000 per year for UG years 1-3; 20,000 per year for years 4-5 / PG	2026-27 portal opened 1 Jun 2026; one listing shows 31 Oct 2026 deadline **[VERIFY]**
Post-matric scholarships	National Scholarship Portal and Karnataka SSP	SC/ST/OBC/minority categories, income limits	Varies	Typically Jul-Nov **[VERIFY]**

Note: CSSS applies to students scoring in the top 20 percentile of Class 12; it may not apply to ITI / diploma students entering after Class 10. **[VERIFY]**

6. Courses, training and institutions

Course / programme	Duration	Entry	Note
Diploma in Mechanical / Civil / Electrical / Electronics / Computer Science / Automobile / Architecture / Mining etc.	3 yr	SSLC; DTE counselling	Technical diploma, lateral entry option
ITI trades (Electrician, Fitter, Welder, Turner, Machinist, COPA, Plumber, Carpenter etc.)	6 mo - 2 yr	Class 8 or 10 by trade	NCVT / NTC certificate
D.Pharm (after Class 12)	2 yr	PCB / PCM **[VERIFY]**	See Health and Allied document
BE / BTech (lateral entry)	3 yr	DCET	Engineering degree after diploma

Examples of institutions: Government Polytechnics across Karnataka (for example in Bengaluru, Mangaluru, Davangere), CITD / Government Tool Room and Training Centre (GTTC) **[VERIFY]**, Government and private ITIs, KSDC training centres. **[VERIFY availability, fees and approvals]**

7. Career outcomes

Path	Job roles	Higher study / growth
Technician and supervisor	Electrician, machinist, fitter, welder, CNC operator, maintenance technician	Skill upgrades, apprenticeship, NSQF levels
Junior engineer	Junior Engineer in PSUs, state departments, railways, private industry	BE via lateral entry, GATE
Skilled trades and construction	Plumber, carpenter, mason, solar and EV installer	Contractor licence, entrepreneurship
Entrepreneurship	Workshop owner, service centre, fabrication unit	MSME and skill loans

8. Related umbrellas this one connects to

- Skills and Entrepreneurship: short-term courses, PMKVY and start-up support (see Skills and Entrepreneurship document).
- Science PCM: engineering degree via Class 11-12 or lateral entry (see Science PCM document).
- Defence: Agniveer Tradesman and technical entries (see Defence document).
- Health and Allied: D.Pharm and paramedical diplomas (see Health and Allied document).
- Agriculture: agricultural diploma and machinery trades (see Agriculture document).

9. Verification log (17 Oct 2026)

What the web check found. Confidence reflects secondary sources only.

Item	Finding	Confidence
Karnataka diploma admission	DTE coordinates; SSLC-based; merit vs entrance conflicts; 35% in Science and Maths (one listing)	Low-Medium
DCET 2026	Apps 17 Mar - 3 Apr; exam 23 May (aggregator)	Low
ITI eligibility and process	Class 8 / Class 10 by trade; 6 months to 2 years; 130+ trades; NTC via AITT; Karnataka notification not found	Medium for general rules; Low for Karnataka
Apprenticeship	ITI is the standard route; no official scheme details found	Low

Still to verify on official sites

- DTE Karnataka 2026-27 diploma admission notification (eligibility, merit rules, seat matrix); DCET 2027 details and eligibility.
- Karnataka ITI trades, eligibility and admission process; NAPS and apprenticeship details.
- Official sources: dtetech.karnataka.gov.in, cetonline.karnataka.gov.in, skilldevelopment.karnataka.gov.in, ncvtmis.gov.in, apprenticeshipindia.gov.in.
- Dates shift every year. Replace "tentative month" with the official notification date before publishing.

ONE-PAGE GIST: Polytechnic and ITI

Draft - verify all **[VERIFY]** items in the detailed version before publishing.

Stage	What to do
Class 6-10	Hands-on skills, Maths and Science basics; tinkering and workshops
After Class 8	ITI trades that accept Class 8 pass (Welder, Plumber, Carpenter, Mason)
After Class 10	Polytechnic diploma (3 yr) or ITI (6 mo - 2 yr)
Diploma to degree	Lateral entry to BE 2nd year via DCET
After ITI	NTC certificate; apprenticeship at PSUs, railways, industry
Career	Technician, junior engineer, supervisor, entrepreneur

Exam / route	When	Pattern / key point	Marking
Diploma (DTE Karnataka)	After SSLC results	3-year diploma; counselling	Check notification
DCET	Yearly (2026: 23 May) **[VERIFY]**	Lateral entry to BE 2nd year	Check notification
ITI admission	After Class 8 / 10	By trade; NCVT / NTC	Merit-based
AITT	End of ITI course	All India Trade Test by NCVT	Check NCVT

Key rules: ITI trades accept Class 8 or Class 10 by trade; diploma needs SSLC; lateral entry keeps the degree option open. **[VERIFY]**

Scholarships: NMMS (Class 8), state scholarships via SSP. **[VERIFY]**

Careers: technician, junior engineer, machinist, electrician, supervisor, workshop owner.

Want a personalised roadmap or counselling? Contact LessonPlan, Yelahanka, Bangalore.